

Relation Algebra

§16. Greedy Algorithms

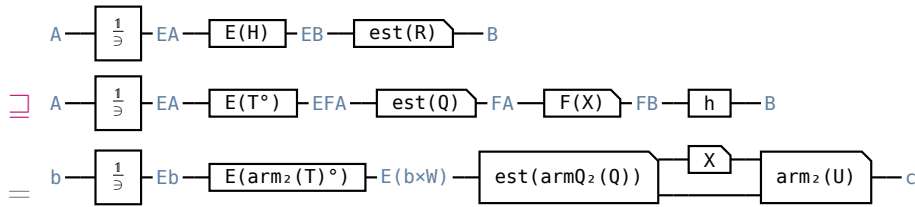
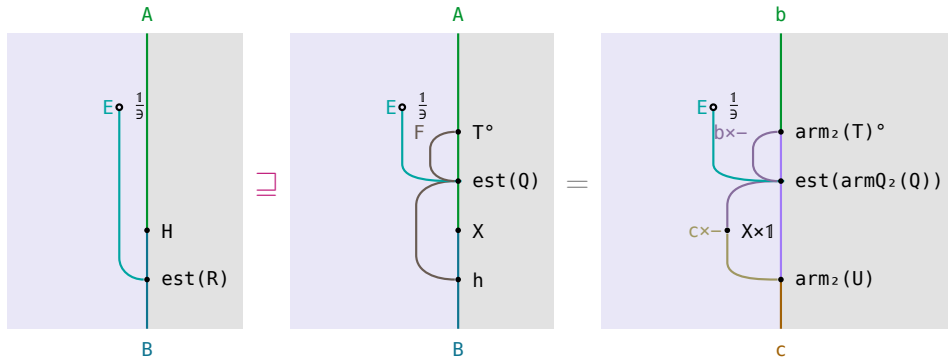
§16.1. Theory

(16.1a)

h, T, R, H, M as in $\triangleq H$; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\exists}$ returns, so that $\frac{T^\circ}{\exists} \text{ est}(Q)$ is entire.

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R) \wedge R R \subseteq R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h) \subseteq \frac{H}{\exists} \text{ est}(R) \quad (16.1b)$$

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

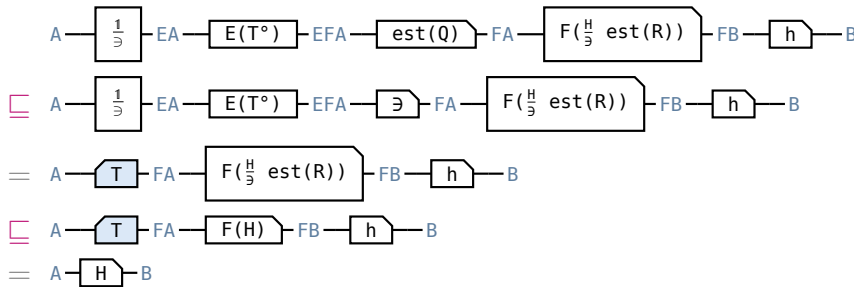
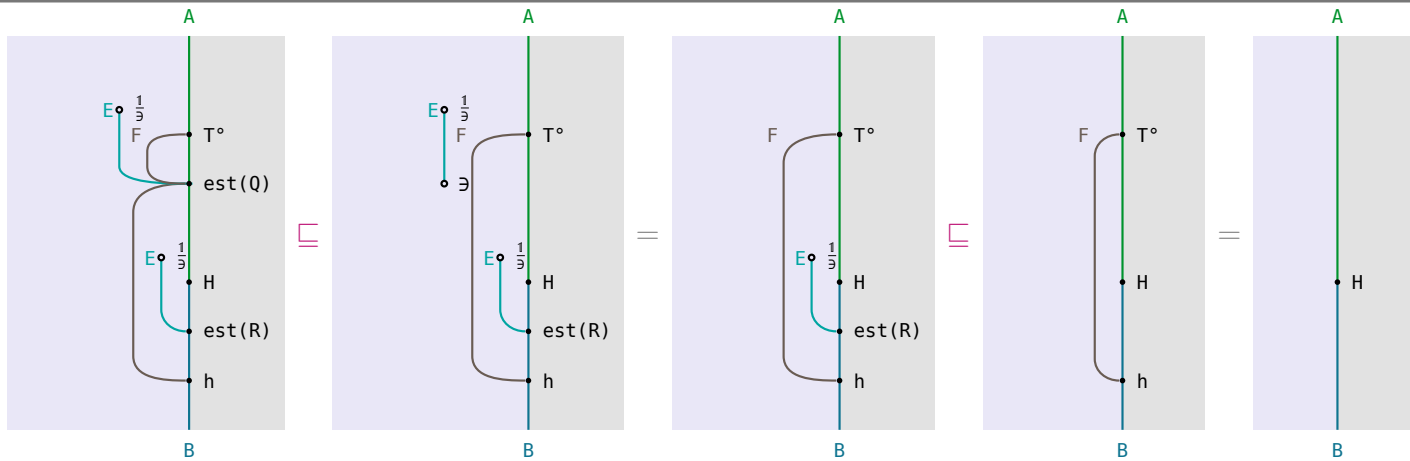


the problem to be solved, $H \triangleq (T^\circ)^\circ(h) \triangleq H, Q$

Proposition 10.1 at $T=[V_1, V_2]$, $h=[U_1, U_2]$, $Q=Q_1+Q_2$, $V_2 V_1^\circ = \emptyset$

$$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \subseteq H \quad (16.1c)$$

taking the input apart in one Q -extreme way, solving the parts by M and assembling by h returns only what H returns



$\text{est}(Q) \subseteq \exists \triangleq \text{est}$

$\frac{T^\circ}{\exists} \exists = T^\circ$ — pow-laws

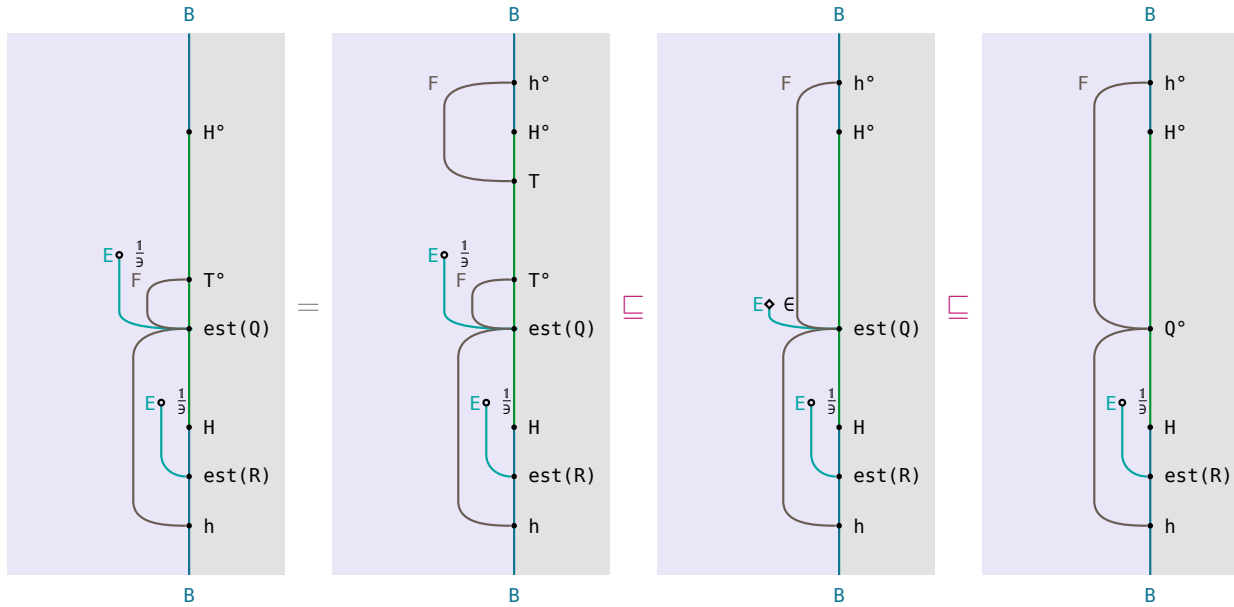
$M \subseteq \frac{H}{\exists} \exists = H$ — est-up

$T^\circ F(H) h = H$: $H \triangleq (T^\circ)^\circ(h)$ and hylo-fix

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H)h = H \wedge QF(H)h \subseteq F(H)hR \Rightarrow H^\circ \frac{T^\circ}{\exists} \quad (16.1d)$$

$$\text{est}(Q)F(\frac{H}{\exists} \text{est}(R))h \subseteq R^\circ$$

H° followed by the greedy body at M is $\subseteq R^\circ$: an answer of the body is never worse than an answer of H to the same input

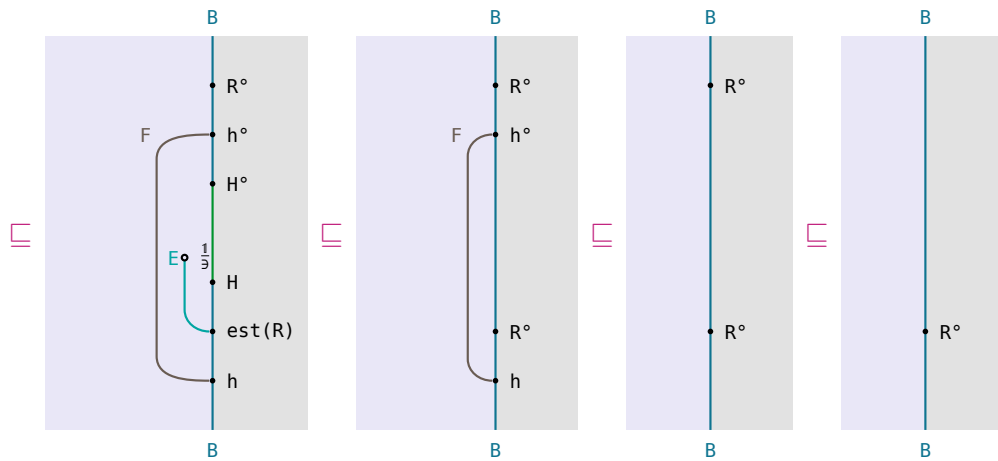


$$\begin{aligned} & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\text{est}(Q)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \\ = & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{T} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\text{est}(Q)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{\epsilon} EFA \xrightarrow{\text{est}(Q)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} Q \xrightarrow{FA} F(\frac{H}{\exists} \text{est}(R)) \xrightarrow{FB} h \xrightarrow{B} \end{aligned}$$

$H^\circ = h^\circ F(H^\circ)T$, the converse of $T^\circ F(H)h = H$ — hylo-fix

$T \frac{T^\circ}{\exists} \subseteq \epsilon$, not a tabulated row: $\frac{T^\circ}{\exists} \ni = T^\circ$ conversed

$\epsilon \text{est}(Q) \subseteq Q^\circ$ — est-up



$$\begin{aligned} \sqsubseteq & B \xrightarrow{R} h \xrightarrow{FB} F(H^\circ) \xrightarrow{FA} F(\frac{H}{\exists} \text{est}(R)) \xrightarrow{FB} h \xrightarrow{B} \\ \sqsubseteq & B \xrightarrow{R} h \xrightarrow{F(R^\circ)} FB \xrightarrow{h} B \\ \sqsubseteq & B \xrightarrow{R} R \xrightarrow{B} \\ \sqsubseteq & B \xrightarrow{R} B \end{aligned}$$

$QF(H)h \subseteq F(H)hR$ conversed — the hypothesis on Q

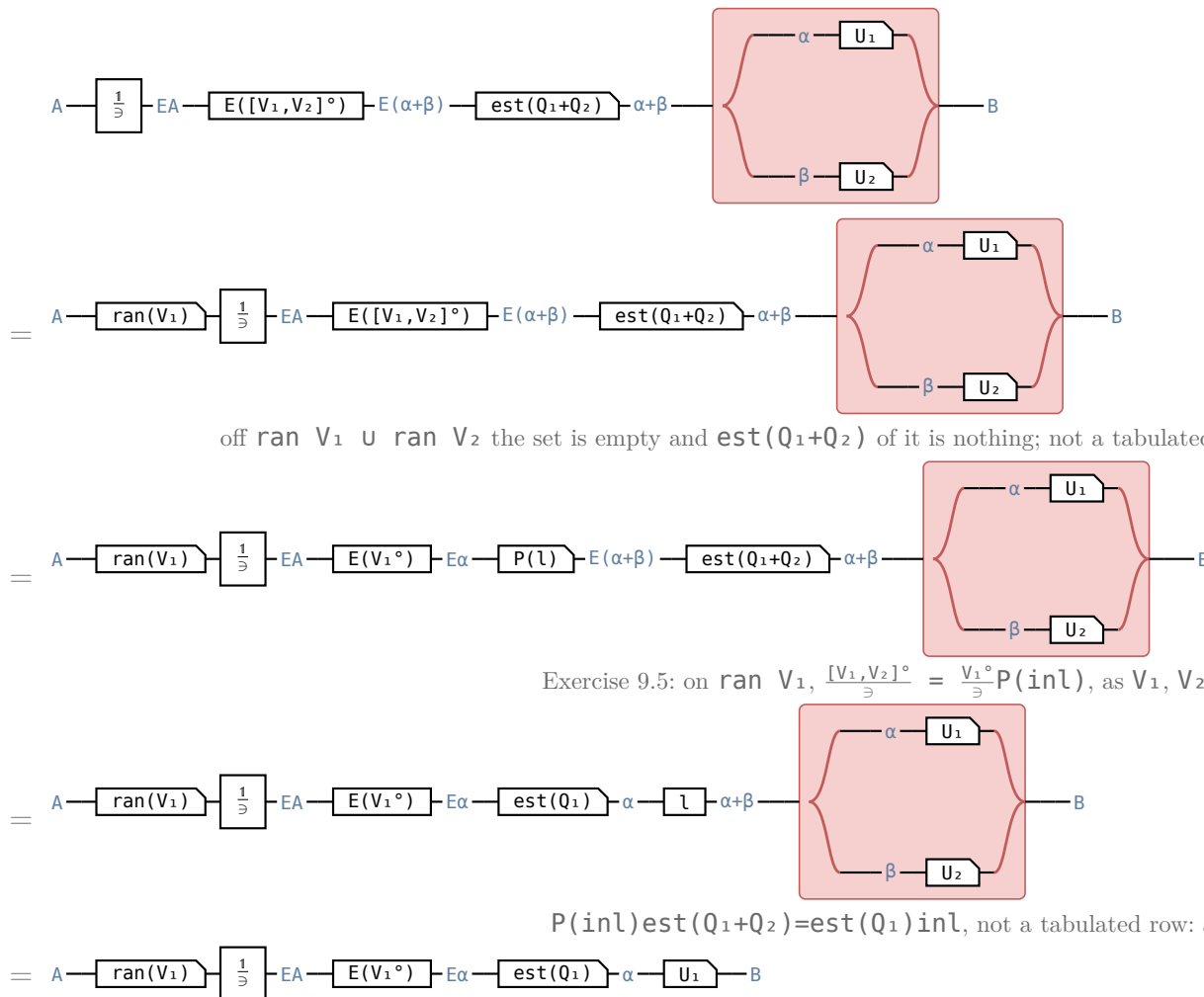
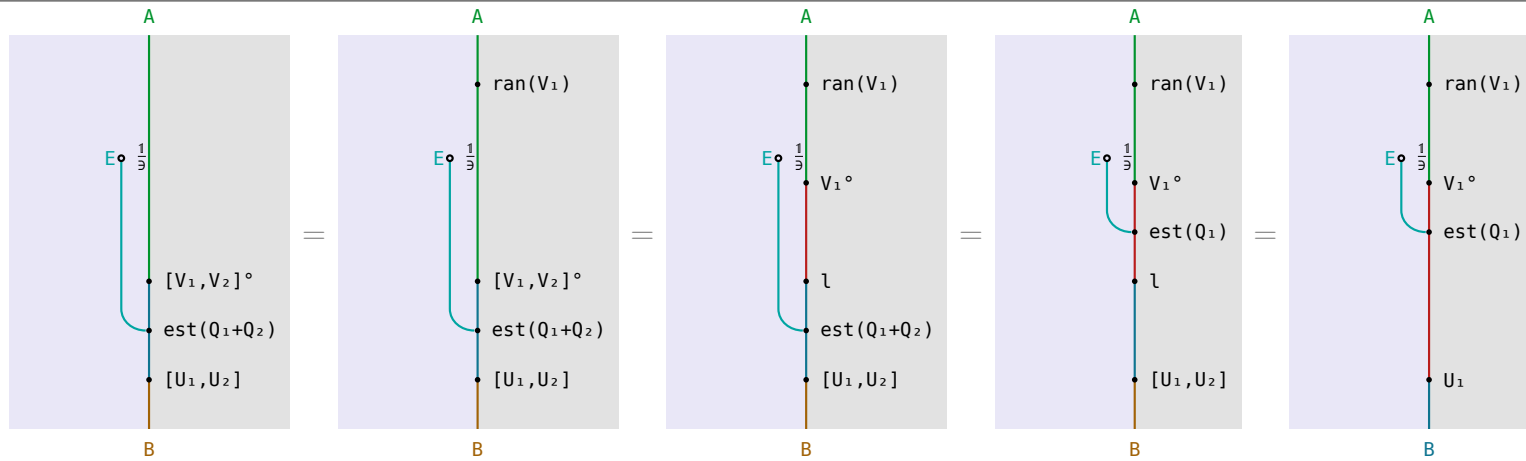
$H^\circ M \subseteq R^\circ$ under F — est-up

$h^\circ F(R^\circ)h \subseteq R^\circ$: h monotonic on R , shunted — the hypothesis on h

R transitive

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1+Q_2)[U_1, U_2] = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{est}(Q_1)U_1 \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{est}(Q_2)U_2 \quad (16.1e)$$

when V_1 and V_2 have disjoint ranges, the greedy step over $[V_1, V_2]$ runs the V_1 step on inputs V_1 reaches and the V_2 step on inputs V_2 reaches



§16.2. The detab-entab problem

(16.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$ over snoc -lists, $\alpha \triangleq [\text{nil}, \text{snoc}]$, $H = \text{detab}^\circ$; $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$, $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$.

$\text{col} \triangleq ([\text{zero}, \text{count}])$, $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c+1)$; TB the tab, BL the blank, NL the newline, tab stops every n columns.

$R \triangleq \text{length} \leq \text{length}^\circ$, U the preorder with $a \cup b \iff a = \text{TB} \vee a = b$, $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$, $Q \triangleq 1 + (V \times U)$.

$\text{unfill } xs$ the shortest prefix of xs with $\text{fill } (\text{unfill } xs) = \text{fill } xs$; tbc the trailing blank count, $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$.

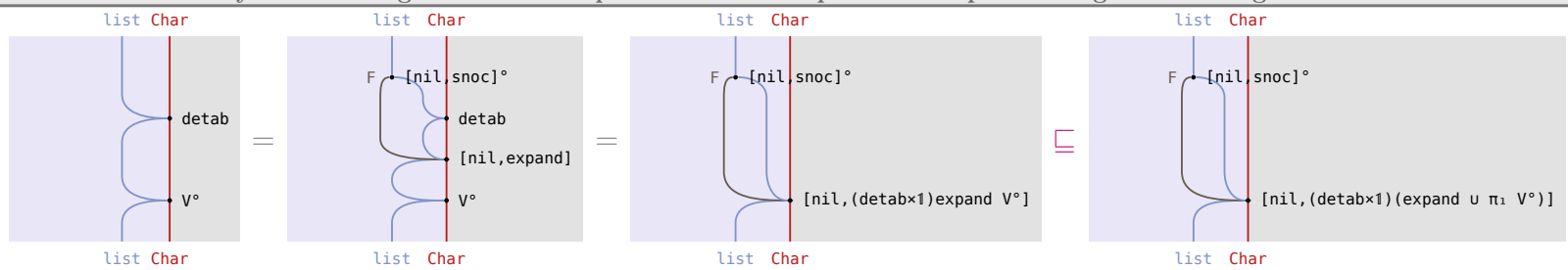
shortening the output of one **expand** step to a **V**-smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ($\pi_1 V^\circ$)



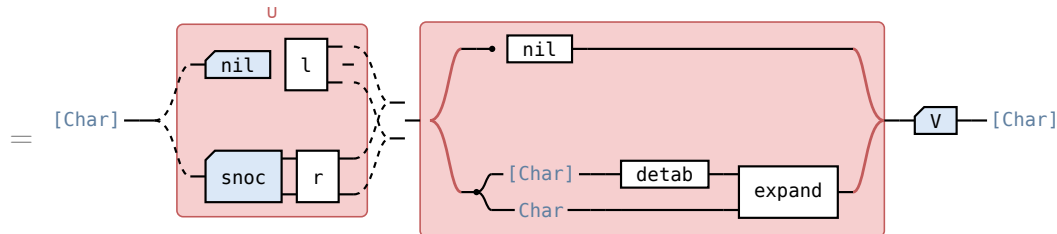
$$0 < n \wedge \text{blank} \neq n1 \Rightarrow \text{detab } V^\circ \in R^\circ \text{detab}$$

(16.2c)

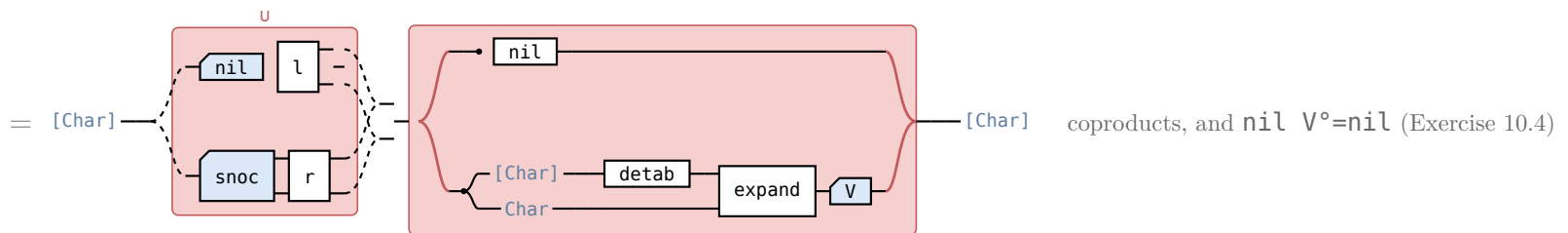
any V -shortening of detab 's output is detab 's output on an input no longer than the given one



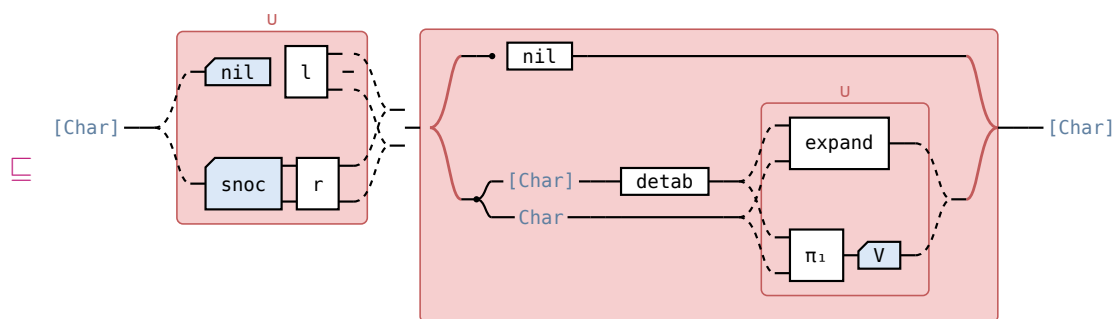
$$[\text{Char}] \xrightarrow{\text{detab}} V \xrightarrow{[Char]}$$



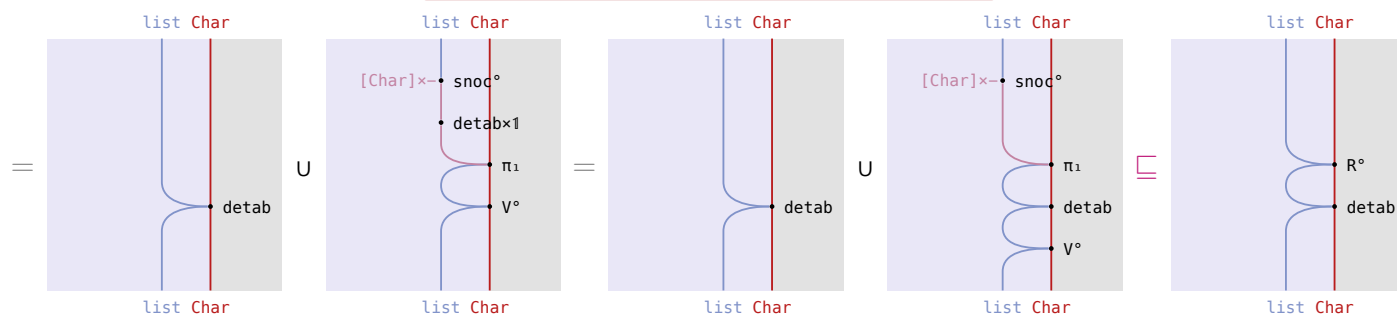
detab is a fold: $\text{detab} = \alpha^\circ F(\text{detab}) [\text{nil}, \text{expand}] \triangleq \text{detab}$



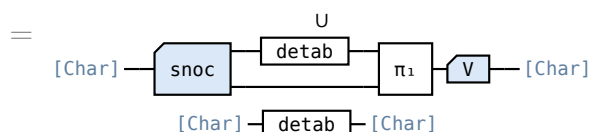
coproducts, and $\text{nil } V^\circ = \text{nil}$ (Exercise 10.4)



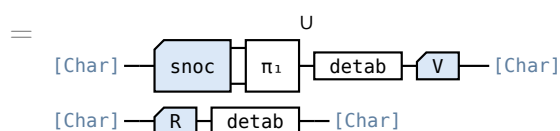
the claim — (16.2b)



$$[\text{Char}] \xrightarrow{\text{detab}} [\text{Char}]$$



distributing U ; the fold again, and the definition of F

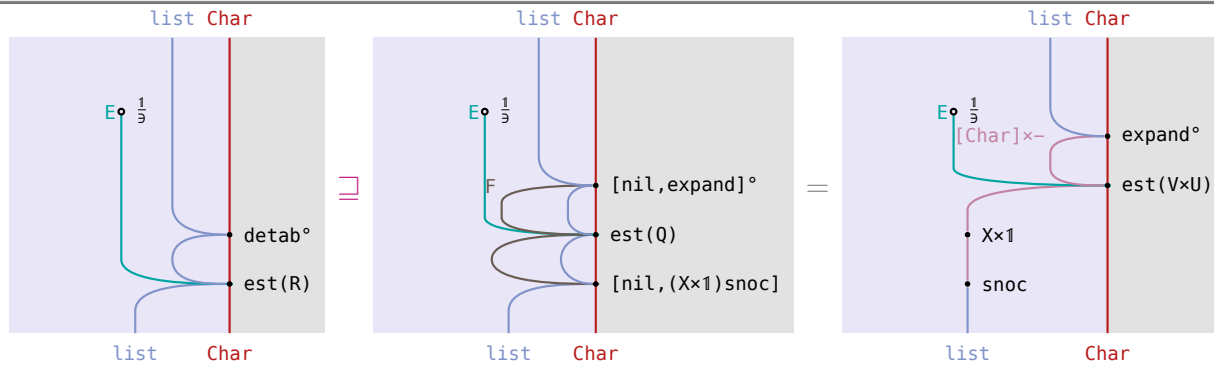


naturality of π_1 : $(\text{detab} \times 1) \pi_1 = \pi_1 \text{ detab}$; $\text{snoc}^\circ \pi_1$ is init

init is inductive, so $X \triangleq \text{detab } V^\circ$, a solution of $X \sqsubseteq \text{detab} u (\text{init } X)$, lies below the greatest one, prefix detab (induction on the input); $\text{prefix} \in R^\circ$

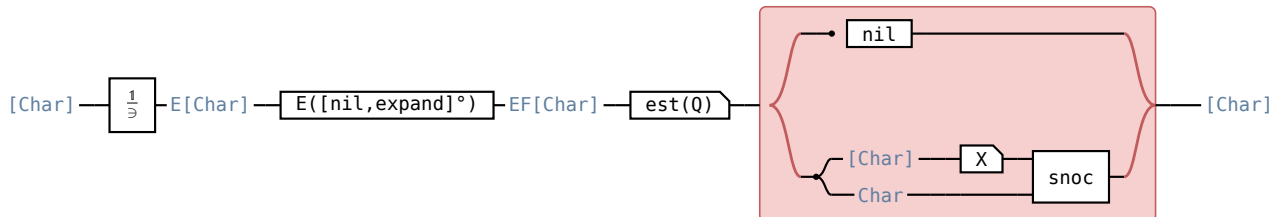
$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R) \quad (16.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and caching the held blanks in for a tab wherever the column reaches a tab stop



$$[\text{Char}] \xrightarrow{\frac{1}{3}} E(\text{detab}^\circ) \rightarrow E[\text{Char}] \rightarrow \text{est}(R) \rightarrow [\text{Char}]$$

the specification — $\triangleq \text{detab}; \text{detab} \text{ entab}=1$ and nothing shorter does



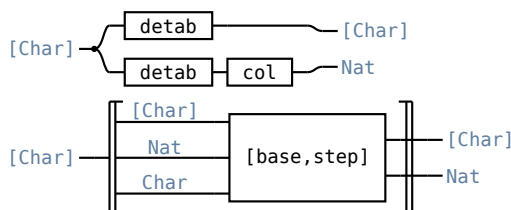
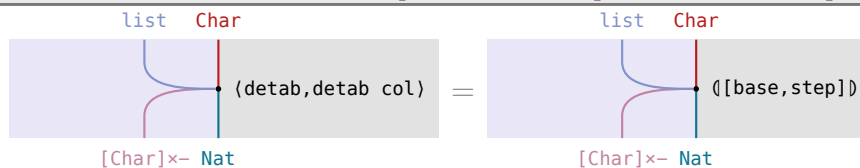
at $Q \triangleq 1 + (V \times U)$: one character of input is decided at each step. $F(T, R) \alpha \sqsubseteq \alpha R$. **detab** prefix $\sqsubseteq R^\circ$ **detab** is FALSE, — at $n=8$, **detab** $[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$, whose prefix $[a, b, c, d, e, \text{BL}, \text{BL}]$ is longer than any input giving it, and **detab** $V^\circ \sqsubseteq R^\circ$ **detab** holds. $\text{expand } V^\circ \sqsubseteq \text{expand } U$ ($\pi_1 V^\circ$) — shortening the output either leaves the last step alone or discards it

$$[\text{Char}] \xrightarrow{\frac{1}{3}} E[\text{Char}] \rightarrow E(\text{expand}^\circ) \rightarrow E([\text{Char}] \times \text{Char}) \rightarrow \text{est}(V \times U) \rightarrow X \rightarrow \text{snoc} \rightarrow [\text{Char}]$$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle [\text{base}, \text{step}] \rangle \quad (16.2e)$$

one pass over the input carries the output so far together with its column



both components are folds over the same input; **step** on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle [\text{base}, \text{step}] \rangle (\text{convert } l) \ r \Leftrightarrow r = \text{loop}(\text{step}, (([]), 0), l) \quad (16.2f)$$

the fold of **[base, step]** over the input converted to a snoc-list is the left loop of **step** from **base**

$$\pi_1(\text{loop}(\text{step}, (([]), c_0), x)) = \text{loop}'((f, g), (c_0, x))$$

Exercise 10.1: when **step** only appends $f(c, a)$ to the output and moves the state by g , the output of the loop is $\text{loop}'(f, g)$, which never carries the output

$$a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a) \quad (16.2g)$$

expand produces $x \# [a]$ from a tab after a string that fills to it, or from x and a

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \ a) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

$$\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x) \quad (16.2h)$$

(10.1): the output of **entab** is that of **unfill** x followed by the trailing blanks held back

$$\text{tbc}[] = 0$$

the empty string has no trailing blanks

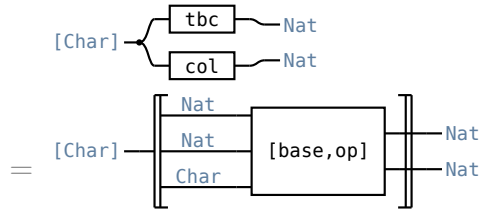
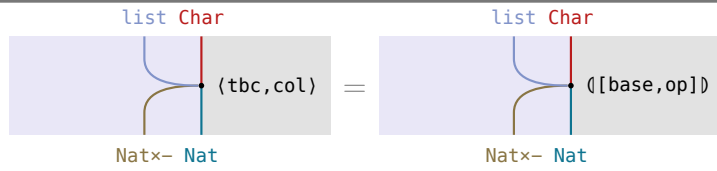
$$\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \ a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq \text{nl} \Rightarrow (\text{tbc}, \text{col}) = ([\text{base}, \text{op}])$$

(16.2i)

one pass counts the held blanks and the column together

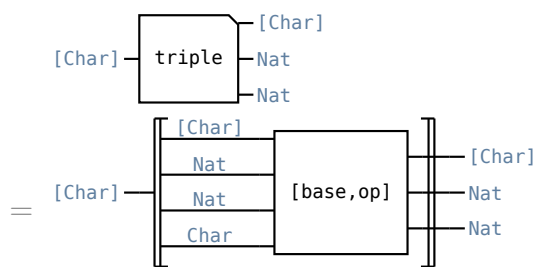
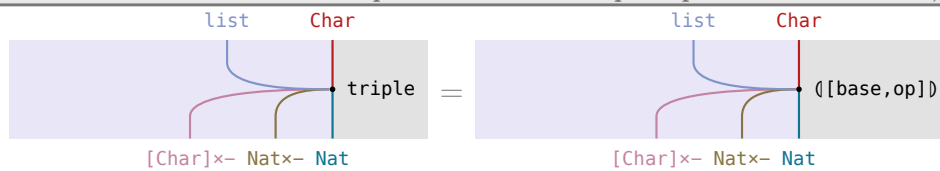


op on (tbc x, col x) is (tbc, col) one character further, from the equations for tbc — (16.2h)

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = ([\text{base}, \text{op}])$$

(16.2j)

one pass carries the output up to the held blanks, their count and the column

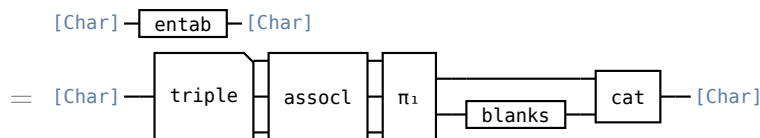
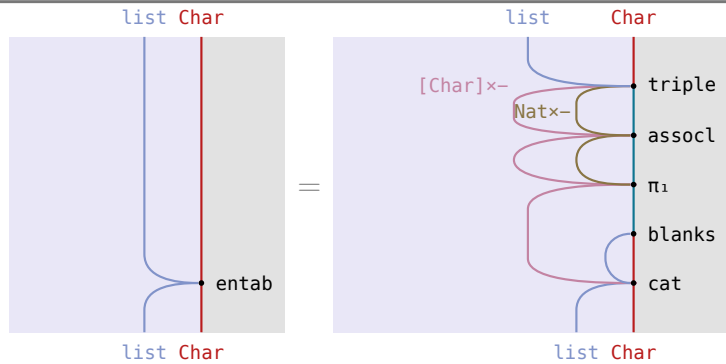


op on triple x is triple one character further: (10.1) where the blanks are released — (16.2h)

$$\text{entab} = \text{triple} \text{ assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

(16.2k)

entab is triple with the held blanks appended to its output



(10.1) — (16.2h)

§16.3. The minimum tardiness problem

(16.3a)

$\text{FX} = 1 + (\text{X} \times \text{Job})$, $\alpha \triangleq [\text{nil}, \text{snoc}]$ on schedules, $\beta \triangleq [\text{nil}, \text{snag}]$ on bags, snag putting a job into a bag; $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$, $\text{H} = \text{bagify}^\circ$.

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$ the completion, due and weighting quantities of a job; $\text{penalty}(\text{xs}, j) = (\text{sum}(\text{list}(\text{ct})(\text{xs})) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j)$.

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$, $\text{cost} [] = 0$, $\text{cost} (\text{xs} \# [j]) = \text{bmax} (\text{cost xs}, \text{penalty} (\text{xs}, j))$, $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$.

$\text{perm} \triangleq \text{bagify} \text{ bagify}^\circ = ([\text{nil}, \text{add}])$, $\text{add} (\text{xs}, j) = \text{ys} \# [j] \# \text{zs}$ for some $\text{xs} = \text{ys} \# \text{zs}$.

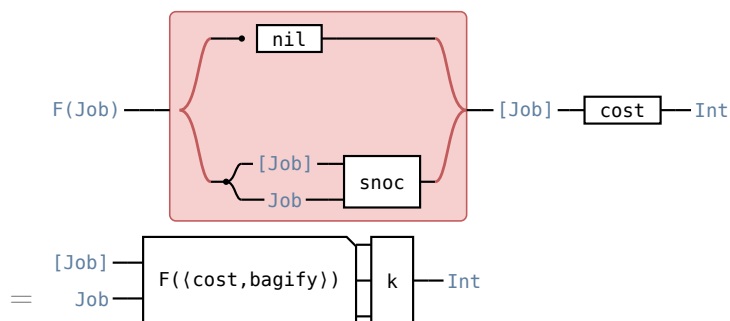
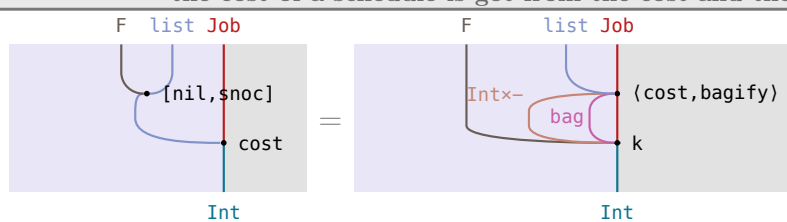
$\text{k} \triangleq [\text{zero}, \text{assocr} (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$, $\text{f} \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$, $\text{Q} \triangleq \text{f} \leq \text{f}^\circ$, $\text{Q}' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$.

$\text{g} \triangleq [\text{zero}, \text{penalty}]$, $\text{m} \triangleq [\text{zero}, \pi_1 \text{ cost}]$ (B&dM's h, renamed as in (16.3k)).

$$[\text{nil}, \text{snoc}] \text{cost} = F(\langle \text{cost}, \text{bagify} \rangle) k$$

(16.3b)

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

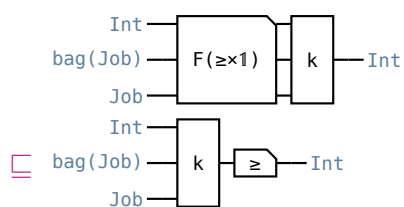
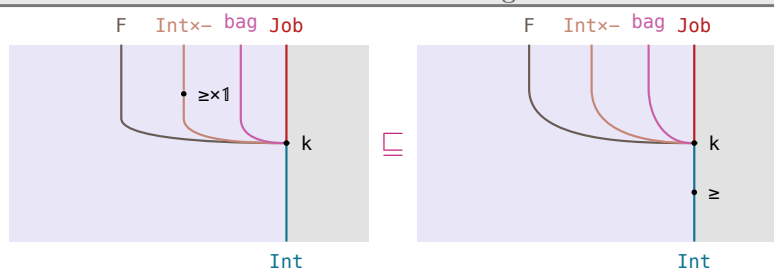


$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$, and $\text{penalty}(xs, j)$ reads only the bag of xs — $\triangleq \text{bagify}$

$$F(\geq \times 1) k \leq k$$

(16.3c)

a larger cost before the last job gives a larger cost after it

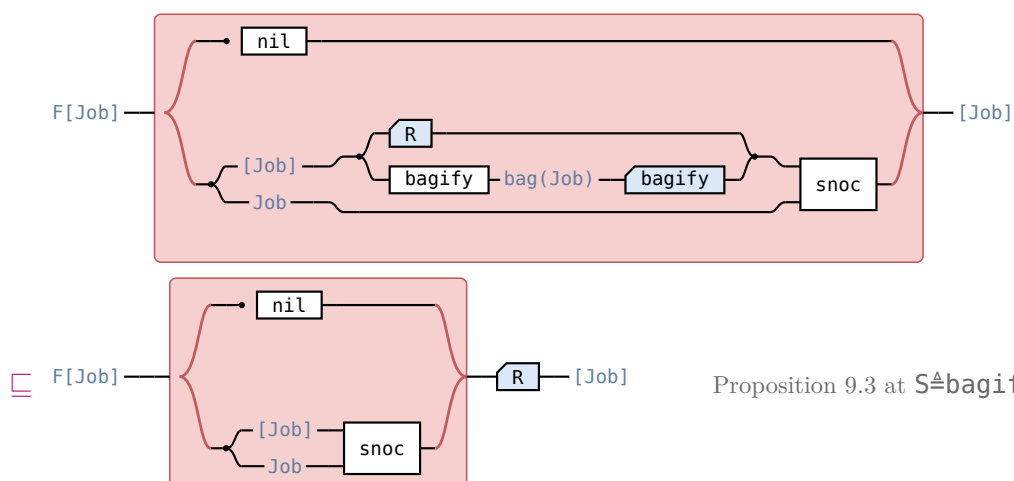
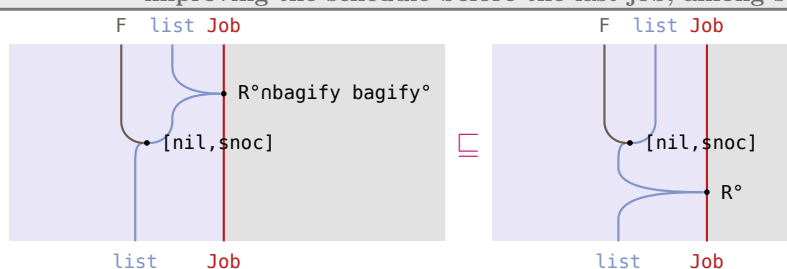


bmax is monotone in its first argument

$$F(R^\circ \text{bagify } \text{bagify}^\circ) [\text{nil}, \text{snoc}] \leq [\text{nil}, \text{snoc}] R^\circ$$

(16.3d)

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule

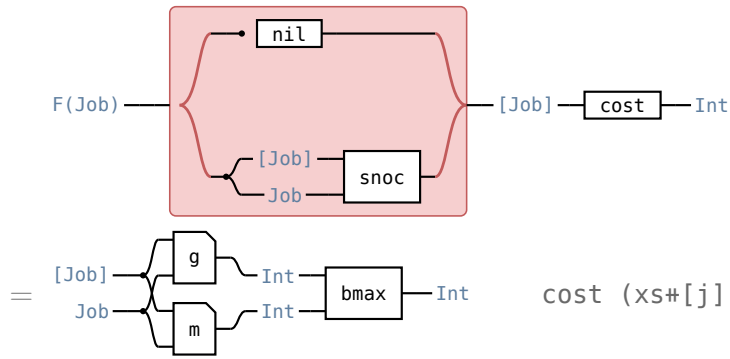
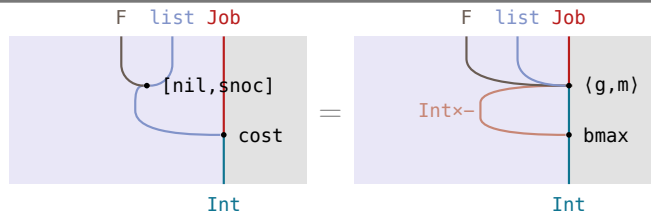


Proposition 9.3 at $S \triangleq \text{bagify}$, $\leq \triangleq$ — dp-context-mono, with (16.3b) and (16.3c)

$$[\text{nil}, \text{snoc}] \text{cost} = \langle g, m \rangle \text{bmax}$$

(16.3e)

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

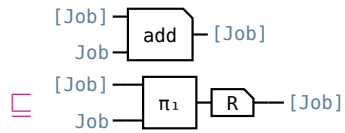
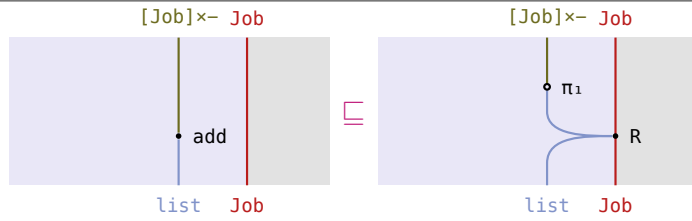


$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ and bmax commutes $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R$$

(16.3f)

putting one more job anywhere into a schedule never lowers its cost

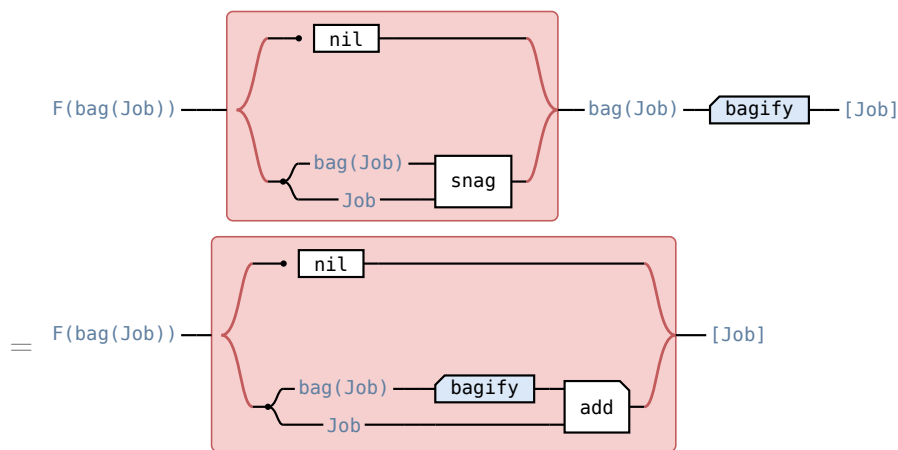
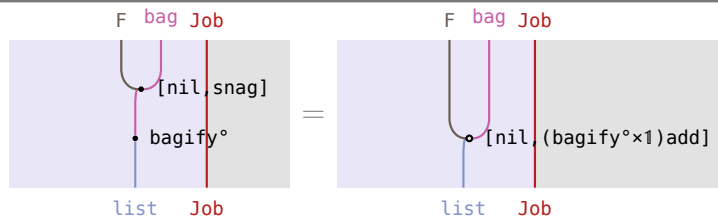


induction on where j goes: every job after it starts later, so no penalty falls

$$[\text{nil}, \text{snag}] \text{bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{add}]$$

(16.3g)

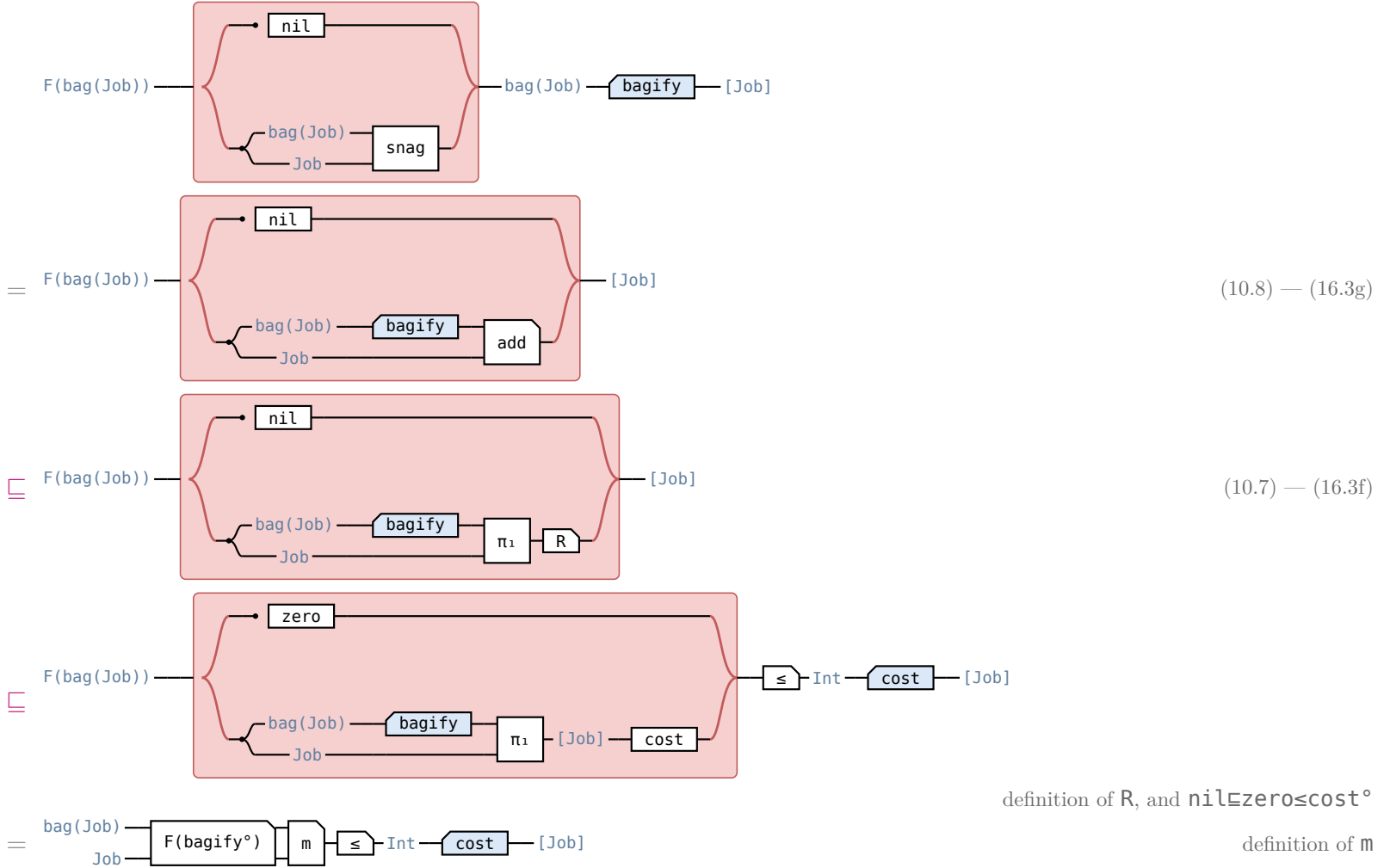
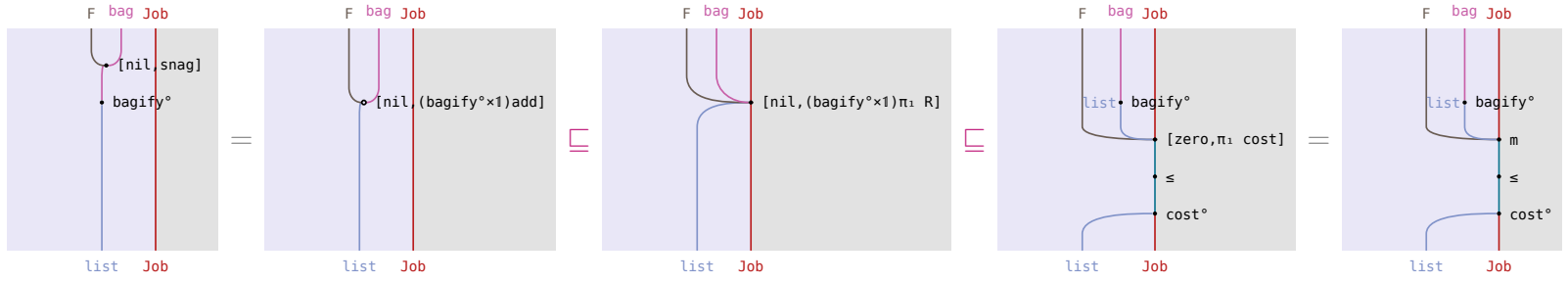
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



take out the last copy of j and put it back $\triangleq \text{bagify}$

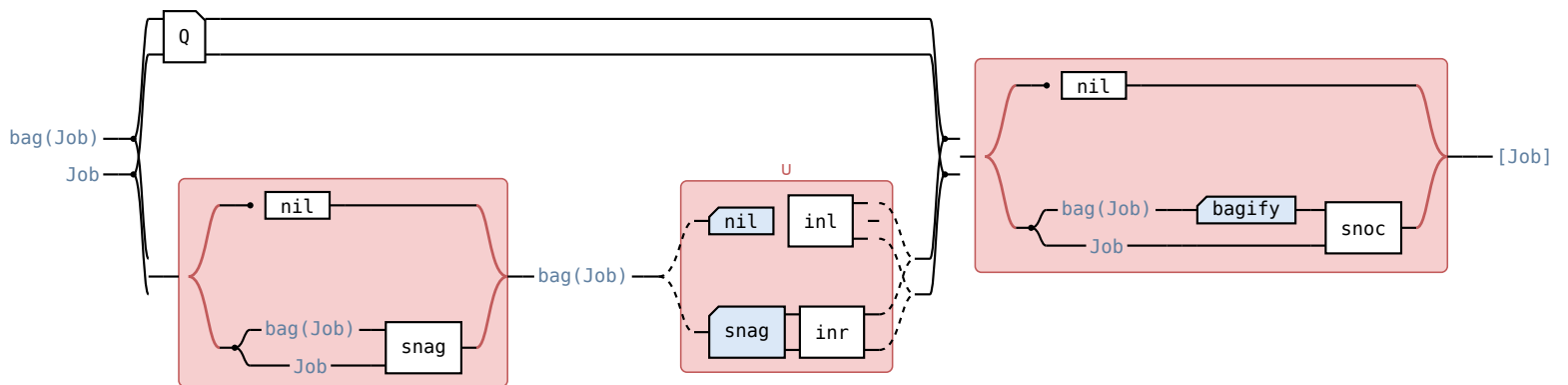
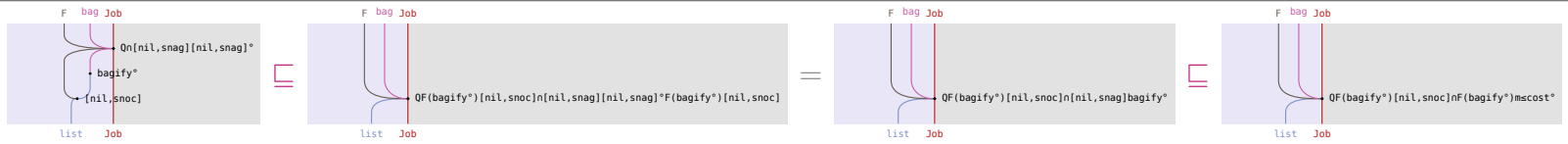
$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow [\text{nil}, \text{snag}] \text{bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (16.3h)$$

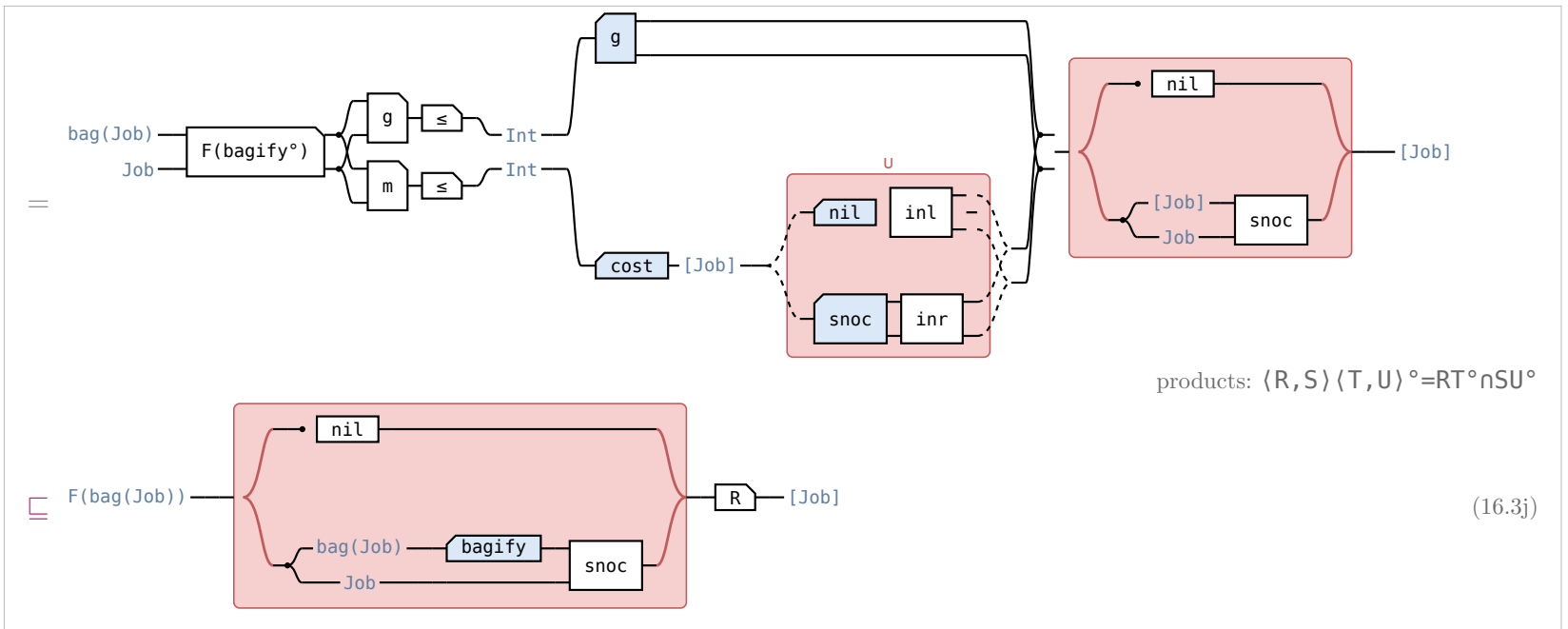
an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job



$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (Qn[\text{nil}, \text{snag}][\text{nil}, \text{snag}]^\circ) F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] \sqsubseteq F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] R \quad (16.3i)$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag

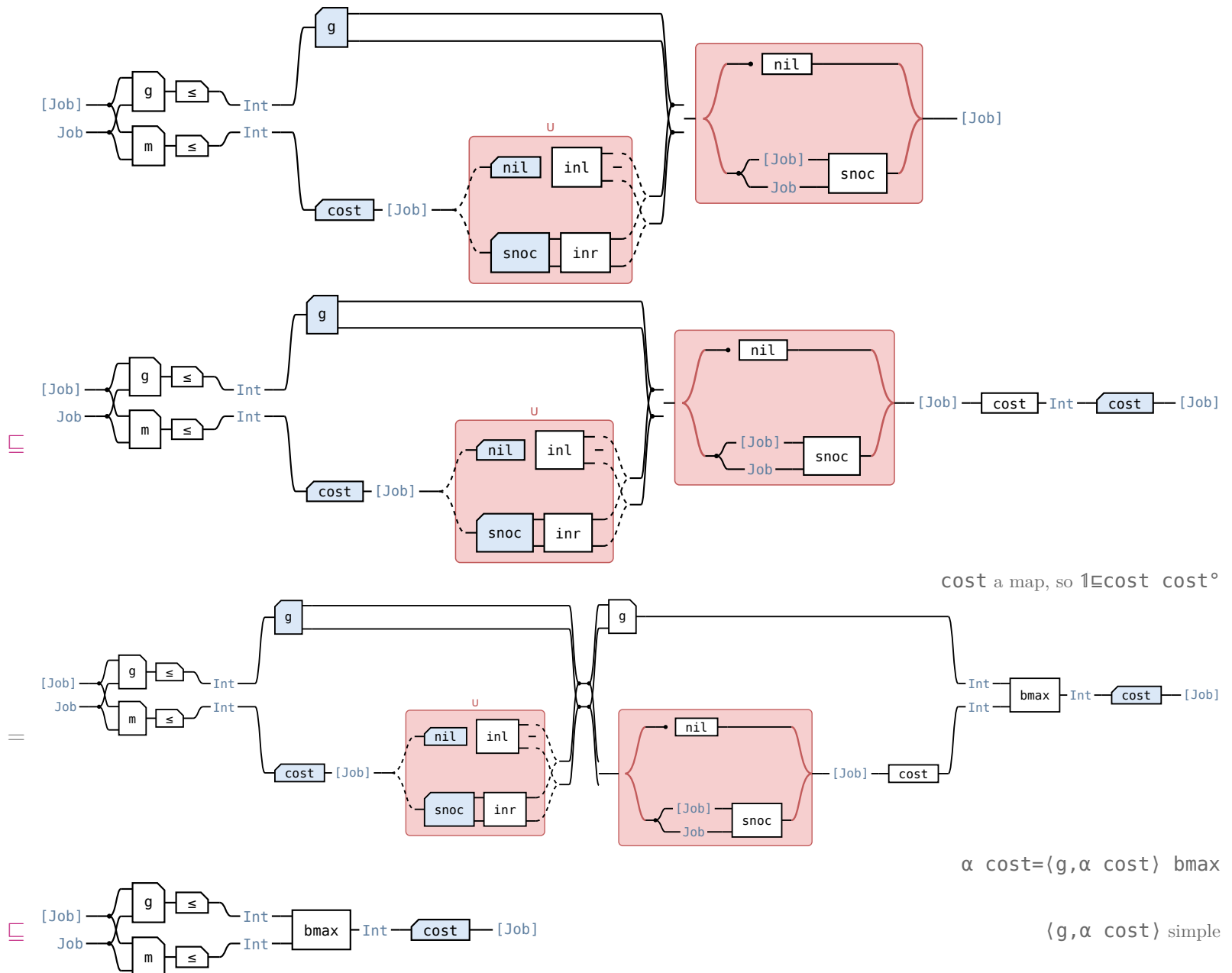
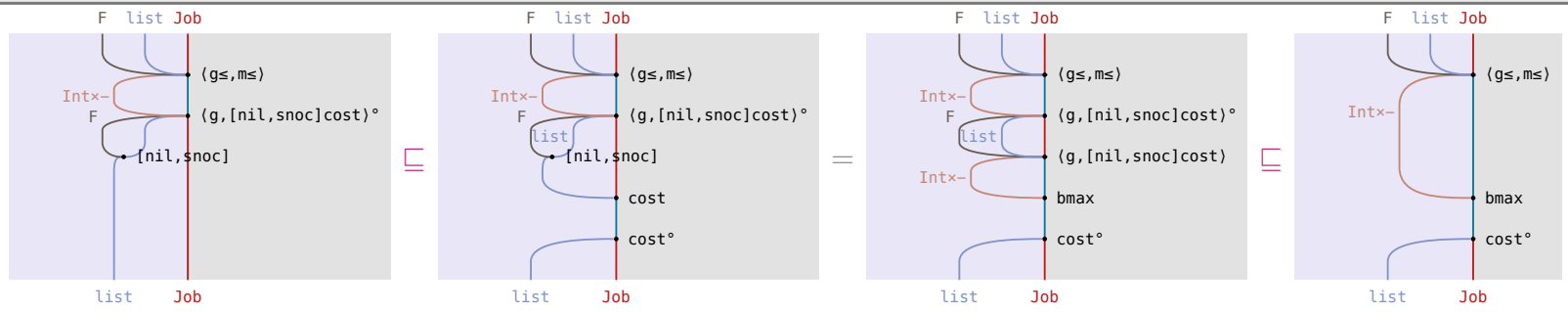


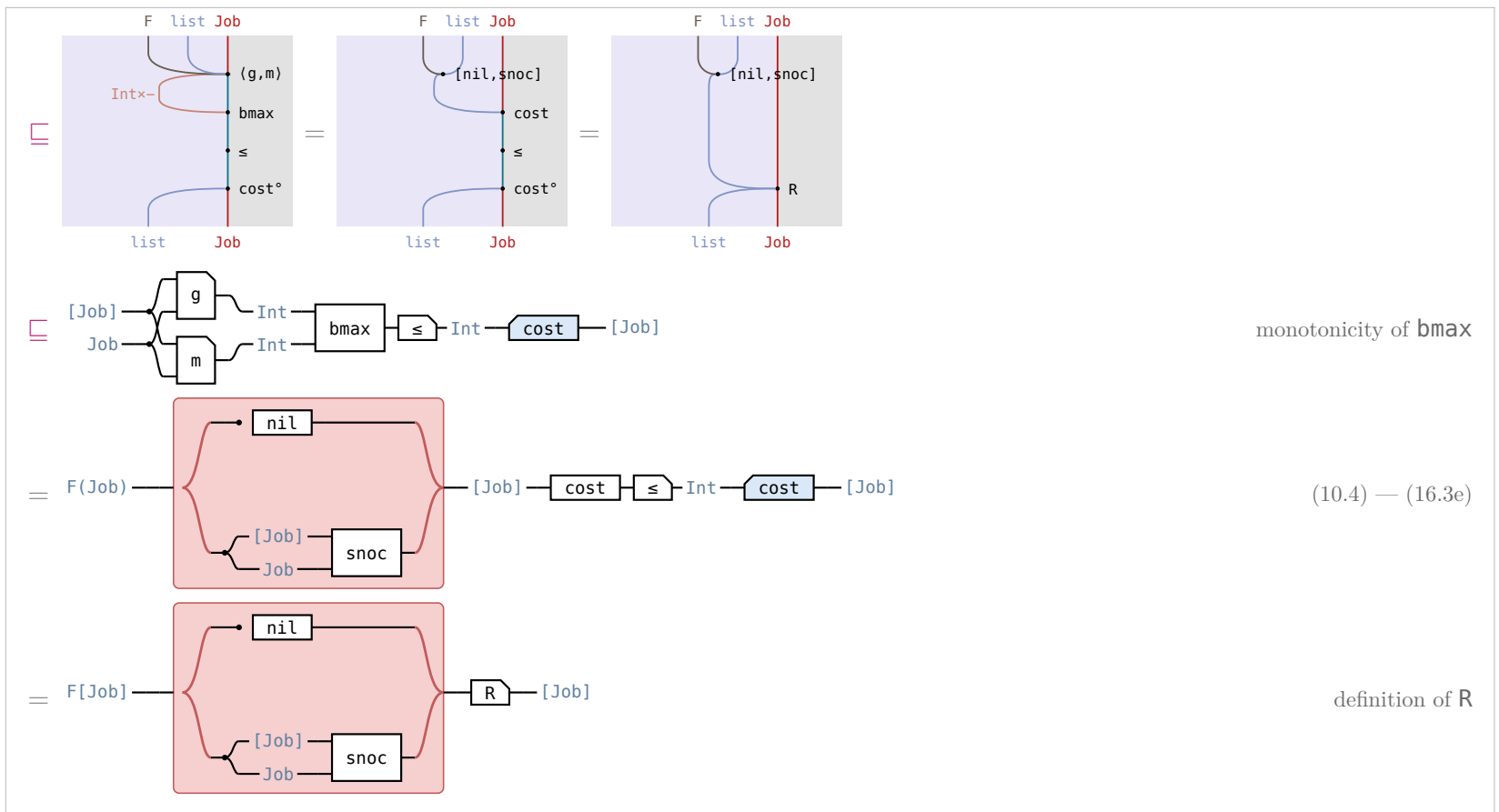


$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R$$

(16.3j)

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job,
 gives a schedule costing at most the whole

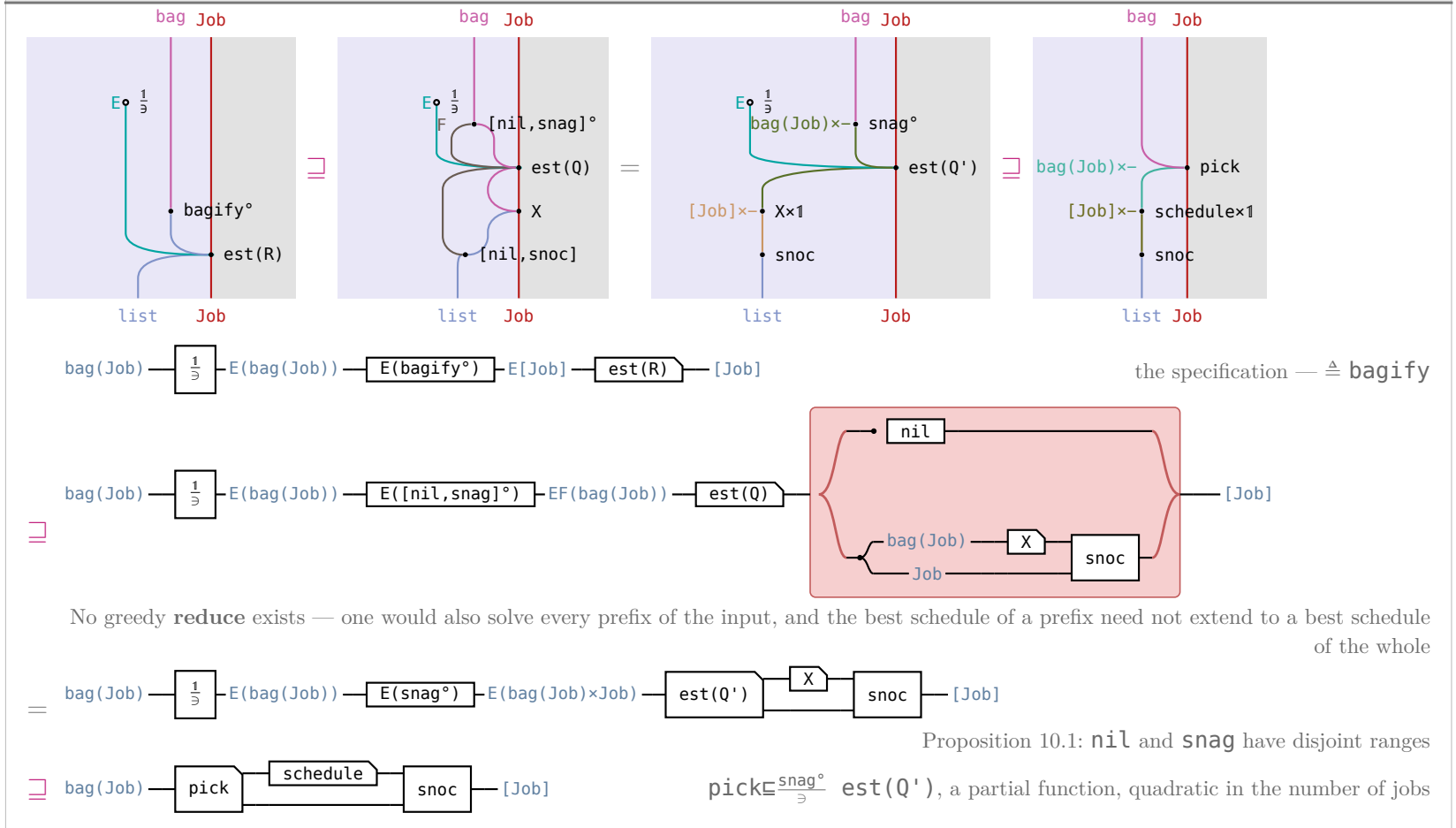




$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq_{\frac{\text{bagify}^\circ}{3}} \text{est}(R), \quad (16.3k)$$

schedule = (null-nil, pick (schedule × 1) snoc)

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way



§16.4. The TeX problem

(16.4a)

intern $\triangleq$ val round : Decimal $\rightarrow$ [0,2¹⁶), val $\triangleq$ Q[zero,shift], shift (d,r)=(d+r)/10, round r rounds 2¹⁶r to the nearest integer: round r=n $\Leftrightarrow$ 2n-1<2¹⁷r<2n+1.

interval n=((2n-1)/2¹⁷,(2n+1)/2¹⁷), r inrange (a,b) $\Leftrightarrow$ a<r<b, round $\circ$ =interval inrange, R $\triangleq$ length $\leq$ length $\circ$.

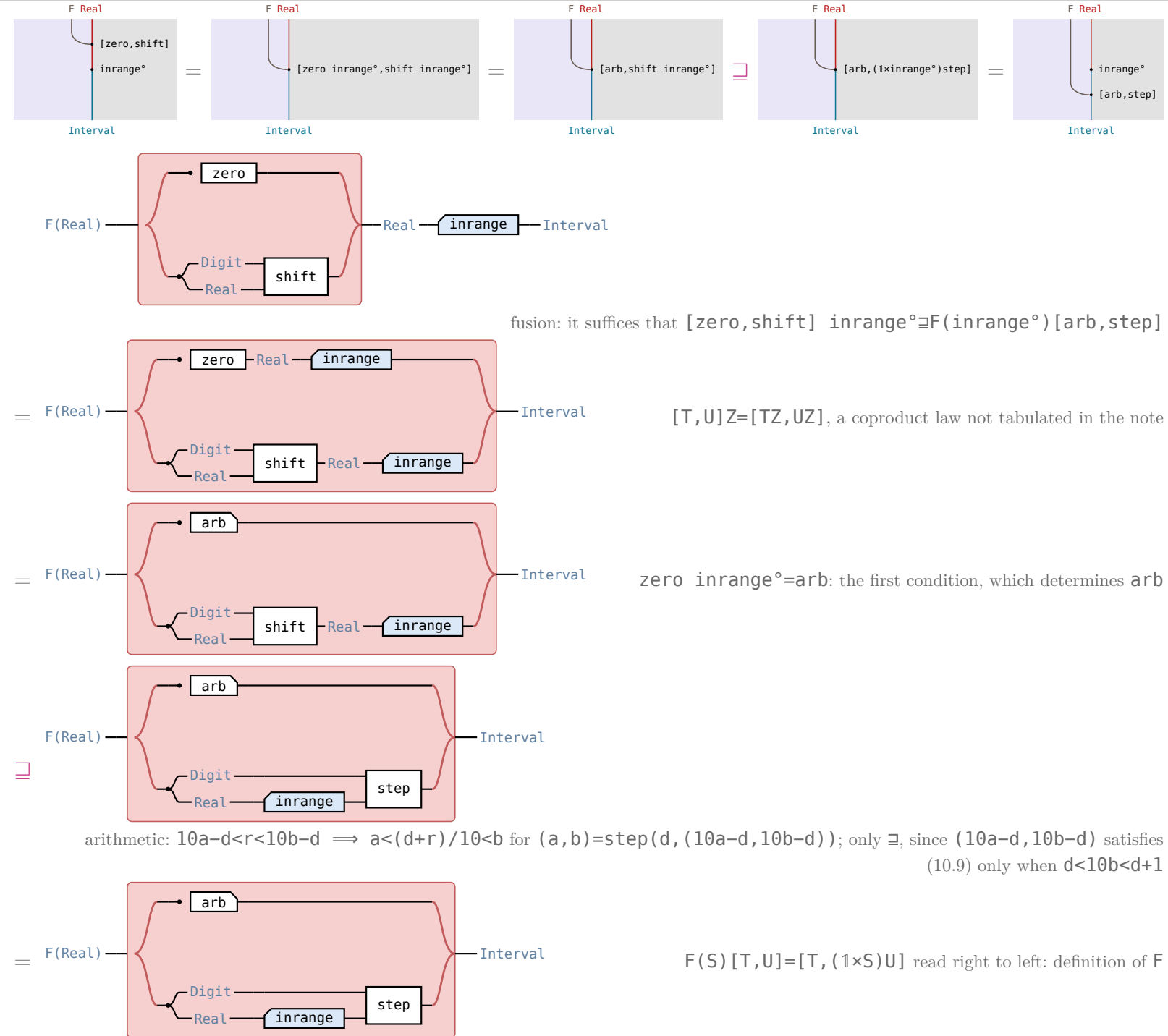
Interval the pairs (a,b) with Legal(a,b) $\Leftrightarrow$ 0<b<1 and a<b (10.9); step_legal is what types step [arb,step] : 1+(Digit $\times$ Interval) $\rightarrow$ Interval, step (d,(a,b))=((d+a)/10,(d+b)/10).

FX=1+(Digit $\times$ X), $\alpha\triangleq$ [nil,cons], H $\triangleq$ Q[arb,step] $\circ$, ! : Digit $\times$ Interval $\rightarrow$ 1, Q $\triangleq$ (l! $\circ$ r) $\cup$ 1 l, r are coprod-laws's injections into FX=1+(Digit $\times$ X), so l : 1 $\rightarrow$ FX and r : Digit $\times$ X $\rightarrow$ FX, w $\triangleq$ 2¹⁷.

Q[arb,step]Eval inrange $\circ$

(16.4b)

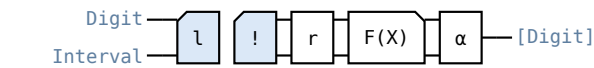
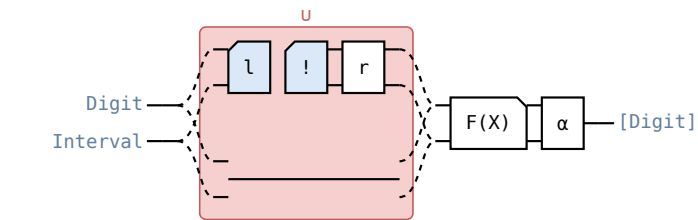
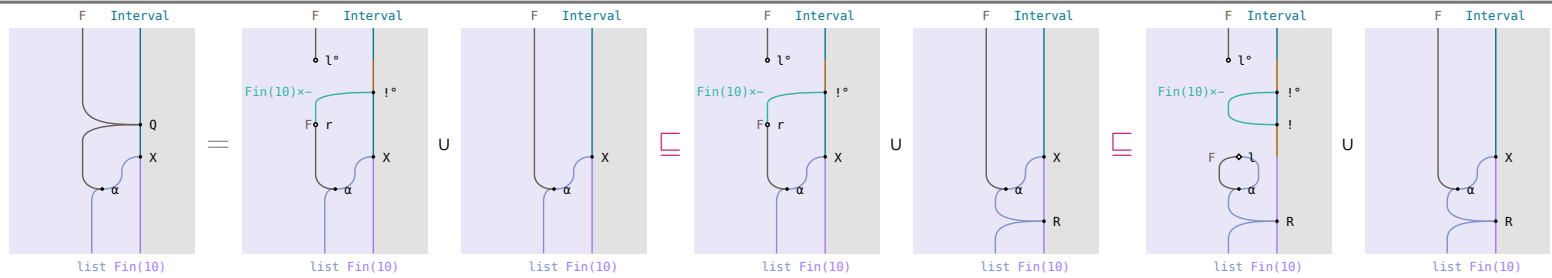
every interval got by folding a decimal's digits with [arb,step] has the decimal's value strictly inside it



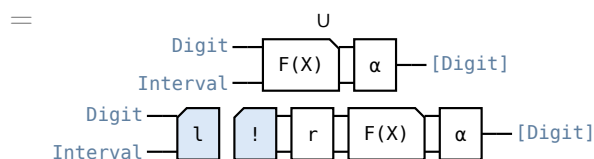
$$QF(X)\alpha \sqsubseteq F(X)\alpha R$$

(16.4c)

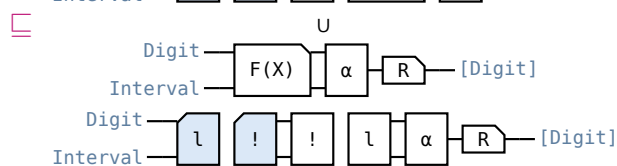
choosing by Q among the one-step decompositions before building with $F(X)$ and α yields only decimals that are R -related to one built without choosing



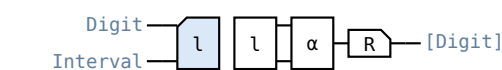
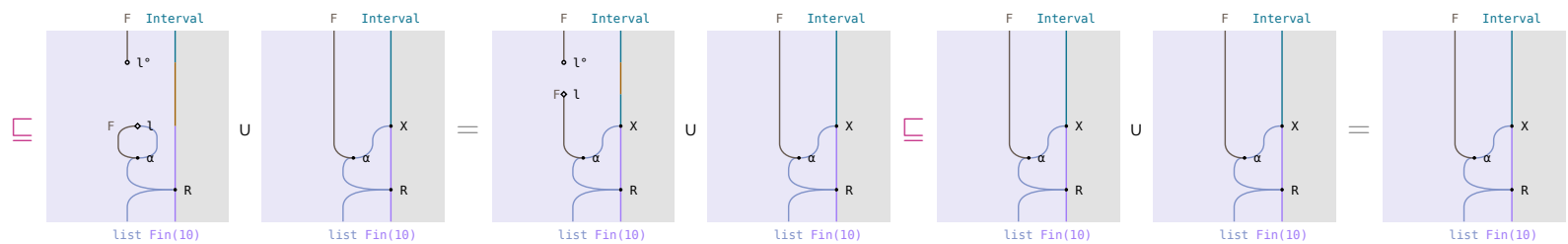
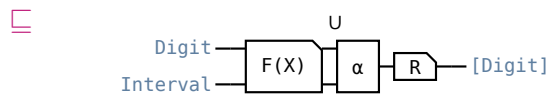
definition of Q ; composition distributes over U



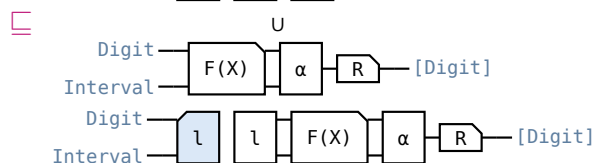
R is reflexive



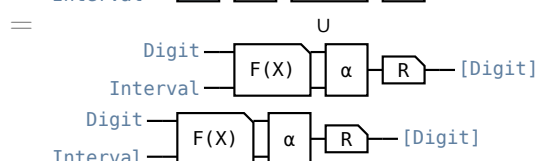
$r F(X) \alpha \sqsubseteq l \alpha R: l \alpha = \text{nil}, \text{ and } \text{length}(\text{nil})=0 \text{ is at most any length}$



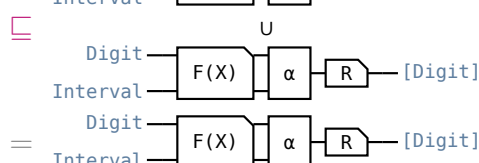
$! \circ ! \sqsubseteq 1$ on 1 : universal property of $!$



$l F(X) = l$: definition of F



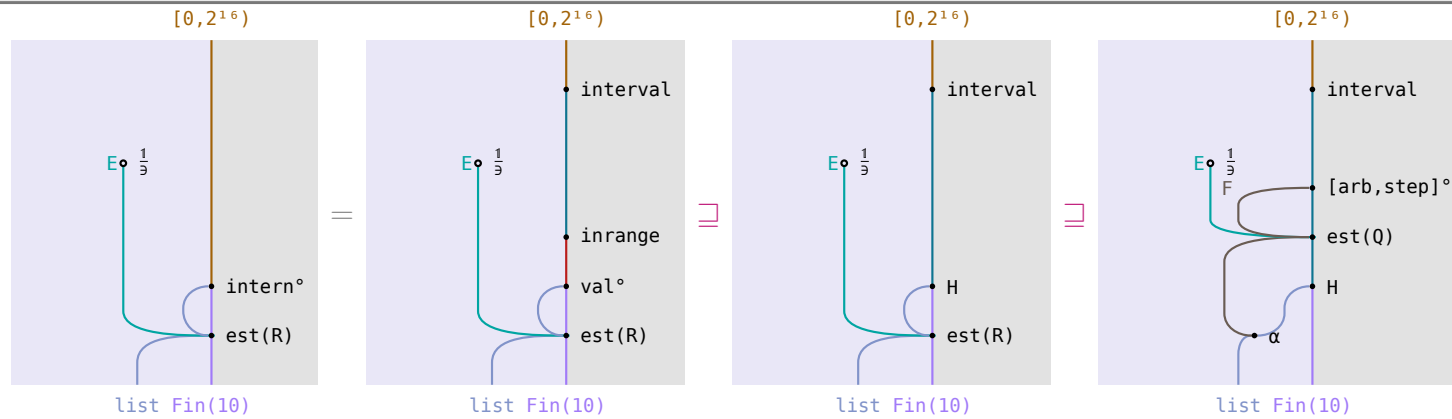
l is simple: $l \circ l \sqsubseteq 1$



U is idempotent

$$\text{interval}(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) , \text{extern}(n)=f(2n-1, 2n+1) \quad (16.4d)$$

a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



$$\begin{aligned}
 & [0, 2^{-16}) \xrightarrow{\frac{1}{3}} E([0, 2^{-16})) \xrightarrow{E(\text{intern}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \quad \text{the specification } \triangleq \text{intern} \\
 = & [0, 2^{-16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(\text{inrange val}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 & \text{round}^\circ \text{ is not a map, but interval is, so it comes out of the transpose} \\
 \sqsubseteq & [0, 2^{-16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(H)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 & \text{the type restriction (10.9): a shortest decimal H gives an interval is a shortest one among all decimals inside it, and is inside it by (16.4b)} \\
 \sqsubseteq & [0, 2^{-16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E([\text{arb}, \text{step}]^\circ)} EF(\text{Interval}) \xrightarrow{\text{est}(Q)} F(H) \xrightarrow{\alpha} [\text{Digit}] \\
 & \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ returns at most two elements — stop, or take one more digit — and ! nil} \sqsubseteq \text{cons } R^\circ \text{ makes it stop whenever stopping is legal}
 \end{aligned}$$

- $f((a, b), x) \iff a < 0 \wedge x = [] \vee \neg a < 0 \wedge (\exists d. \exists y. d = \text{floor}(10 * b) \wedge f((10 * a - \uparrow d, 10 * b - \uparrow d), y) \wedge x = \text{cons}(d, y))$ (16.4e)
 f gives (a, b) the empty decimal when $a < 0$, and otherwise the digit $d = \lfloor 10b \rfloor$ followed by what it gives $(10a - d, 10b - d)$
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) = f$
the least solution of the greedy recursion of (16.4d) is f
- $d = \text{floor}(10 * b) \wedge e = \text{floor}(10 * b) \implies d = e$
Exercise 10.12: two digits d and e with $0 < 10b - d < 1$ and $0 < 10b - e < 1$ are equal
- $(0 < q - p \wedge q < w) \wedge p < 0 \implies f(p, q) = []$
the program's first clause: $f(p, q) = []$ when $p < 0$
- $(0 < q - p \wedge q < w) \wedge \neg p < 0 \wedge d = 10 * q / w \implies f(p, q) = \text{cons}(d, f(10 * p - w * d, 10 * q - w * d))$
the program's second clause: with $d = (10q) \text{ div } w$, $f(p, q)$ is d followed by $f(10p - w * d, 10q - w * d)$
- $(0 < q - p \wedge q < w) \wedge \text{Reach}(p, q) \implies f((p / w, q / w), f(p, q))$
on the pairs the program reaches from $\text{interval } n$, the integer f at (p, q) is a value of the rational f at $(p/w, q/w)$
- $\text{interval } f = \text{extern}$
 f after interval is the program extern , whose value at n is $f(2n-1, 2n+1)$

§17. Appendix

§17.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(17.1a)

(p, q) tabulates $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$, a map.

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, ^\circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, ^\circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{^\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

§17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{est-711}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(17.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{est-73}} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus (\epsilon \setminus R^\circ)))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

§17.3. Shortest paths on a cylinder, on lists

(17.3a)

definition	type	note
$F(A, X) = A + A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
$L = \text{list}^+$	$\mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
α the initial algebra	$F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$	$\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$: a path is started by one square, or extended by one.
N the n-tuple relator	$\mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$[\text{Nat}] \rightarrow [\text{Nat}]$	The cost of a path, which the cheapest minimises.
setify	$A[n] \rightarrow EA$	$\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.
moves	$A[n] \rightarrow E(A[n])$	$\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.
trans	$E(A[n]) \rightarrow (E A)[n]$	$\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components.
zip	$F(a[n]) \rightarrow (F a)[n]$	$\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$.
$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$F(EA) \rightarrow EFA$	$\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$.
$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{zip } N(\text{cp}(F(A, -), T)E(\alpha))$	$F(NA, N(E(TA))) \rightarrow N(E(TA))$	$\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (17.3.1b).
$\text{paths} \triangleq (\text{gen}) \text{setify union}$	$T(NA) \rightarrow E(TA)$	$\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (17.3.1b)'s four sets: 12 paths, 3 from each entry row.
the specification $\text{paths est}(R)$	$[\text{Nat}[n]] \rightarrow [\text{Nat}]$	A cheapest path from the entry side to the exit side.

§17.3.1. $N(E(L A))$

$$\begin{array}{l} 1 \quad 5 \\ 2 \quad 6 \\ 3 \quad 7 \\ 4 \quad 8 \end{array} \quad [(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$$

(17.3.1a)

$$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$$

(17.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ (\{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \}) \end{aligned}$$

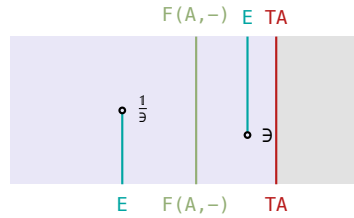
(17.3.1c)

N	the 4-tuple, one component per row of the entry column
E	the set of paths that can start in that row
L	a path is the non-empty list of the squares it crosses

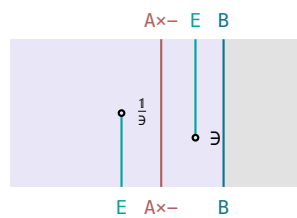
row 1's set has no $[1,7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1,8]$, because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of $\triangleq$ **paths**: component k collects the paths of rows $k-1, k, k+1$.

§17.3.2. The cross product $cp \triangleq \frac{F(1, \exists)}{\exists}$

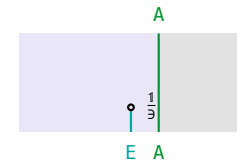
(17.3.2a)



cp on all of $F(A, E(L A))$



the $A \times -$ summand: $\exists$ picks one path, $1\% \exists$ collects the results



the A summand, no E to distribute: $1\% \exists$ alone, $a \mapsto \{a\}$

(17.3.2b)

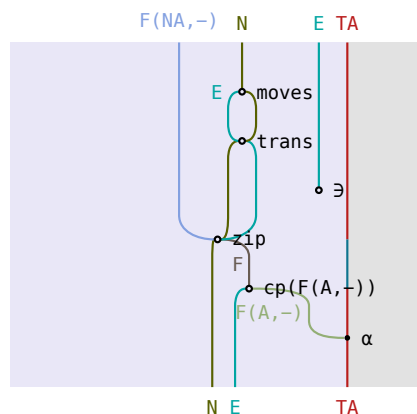
type	note
$F(A, -) = A + A \times -$	one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$
$\exists : E(L A) \rightarrow L A$	the $\exists$ inside $F(1, \exists)$: a set of paths, one of them — $L A$ is one path, A one square
$F(1, \exists) = 1 + 1 \times \exists$	$: A + A \times E(L A) \rightarrow A + A \times L A$, the relator acting on each summand
$\exists : E(A + A \times L A) \rightarrow A + A \times L A$	the $\exists$ under the bar: a set of cells, one of them — a different $\exists$
$cp : A + A \times E(L A) \rightarrow E(A + A \times L A)$	$\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$
$cp = \frac{1}{\exists} E(F(1, \exists))$	what the picture draws: the unit makes the outer E , leaving one $\exists$ — pow-laws

(17.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$ the right summand
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$ 1 keeps the square 1, $\exists$ picks one path
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$ one output per path in w
 $cp(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$ the three of them, collected

§17.3.3. $gen = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(cp P(\alpha))$

(17.3.3a)

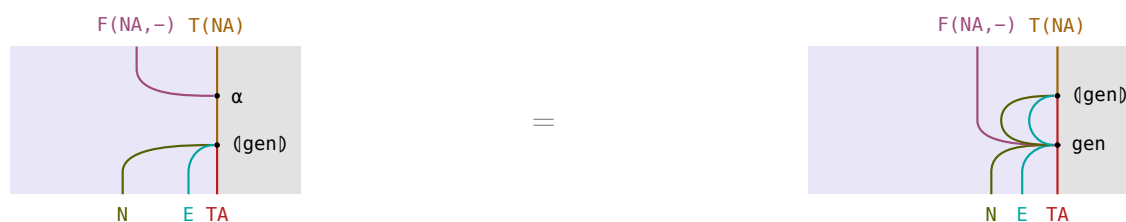


(17.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$: $F(N\ A, N(E(L\ A)))$
 $F(1, \text{moves trans } N(\text{union}))$: 1 keeps the column, the path SETS move
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$
 $= \{(\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})\}$: $E(N(E(L\ A)))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$
 $\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$: $N(E(E(L\ A)))$ row k gets rows k-1, k, k+1
 $N(\text{union})(\text{that})$
 $= (\{[5],[6],[8]\},\{[5],[6],[7]\},$
 $\{[6],[7],[8]\},\{[5],[7],[8]\})$: $N(E(L\ A))$ every path into row k, none dropped
 $\text{zip}(\text{that})$
 $= ((1,\{[5],[6],[8]\}), (2,\{[5],[6],[7]\}),$
 $(3,\{[6],[7],[8]\}), (4,\{[5],[7],[8]\}))$: $N(F(A, E(L\ A)))$ each row: its square, and the paths it may be put in front of
 $N(\text{cp } P(\alpha))(\text{that})$
 $= (\{[1,5],[1,6],[1,8]\},\{[2,5],[2,6],[2,7]\},$
 $\{[3,6],[3,7],[3,8]\},\{[4,5],[4,7],[4,8]\})$: $N(E(L\ A))$ cp pairs the square with each path, α prefixes it: $\alpha(1,[5])=[1,5]$

§17.3.4. gen is an F-algebra; $(\text{gen}): \alpha(\text{gen})=F(1, (\text{gen}))\text{gen}$

(17.3.4a)



α puts the column back on the list, then the fold reads all of it

the fold reads the rest under F , then one gen puts the column in front

(17.3.4b)

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N\ A)$$

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (17.3.1b): the tail folds to $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$, one path per row and each of them one square long, and gen on it is (17.3.3b)'s walk.

§17.3.5. $Q : F(N\ A, N(L\ A)) \rightarrow N(L\ A)$, an $F(N\ A, -)$ -algebra: $Q=F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(17.3.5a)

$Q : F(N\ A, N(L\ A)) \rightarrow N(L\ A)$: the fold's algebra: a new column, one path per row
 $u = ((1,2,3,4),([5],[6],[7],[8]))$: $F(N\ A, N(L\ A))$
 $F(1, \text{moves trans } N(\text{est}(R)))$: 1 keeps the column, the paths move
 $\text{moves}([5],[6],[7],[8])$
 $= (\{[6],[7],[8],[5]\},$
 $([5],[6],[7],[8]),$
 $([8],[5],[6],[7])\}$: $E(N(L\ A))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6],[5],[8]\},\{[7],[6],[5]\},$
 $\{[8],[7],[6]\},\{[5],[8],[7]\})$: $N(E(L\ A))$ row k gets rows k-1, k, k+1
 $N(\text{est}(R))(\text{that})$
 $= ([5],[5],[6],[5])$: $N(L\ A)$ the cheapest into each row – chosen BEFORE the new square
 $\text{zip}(\text{that})$
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$: $N(F(A, L\ A))$ each row: its square, and the one predecessor that survived
 $N(\alpha)(\text{that})$
 $= ([1,5],[2,5],[3,6],[4,5])$: $N(L\ A)$ $\alpha(1,[5])=[1,5]$

$Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (17.3.5b)	
at the last column Q starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (17.3.5c)'s last step folds	
circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$	Hinze–Marsden
$F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$	
$FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$ the fusion condition (17.3.5d) read as a definition,	
$FN(N A) (N (T A)) \text{---} \left\{ \begin{array}{l} N(p_{\circ}) \text{---} \boxed{N(\text{wrapA})} \\ q \text{---} \boxed{\text{pm}} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_{\circ}) \text{---} \boxed{N(\text{consA})} \end{array} \right\} \text{---} N(T A)$ $\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$	

paths $\text{est}(R) \exists (Q)$ setify $\text{est}(R)$ (17.3.5c)	
shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; Q is (17.3.5b)'s algebra.	
circuit	Hinze–Marsden
$T(N\ A) \text{---} \boxed{\text{paths}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$	
$= T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{\text{union}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\triangleq \text{paths at paths}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{P(\text{est}(R))} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $P(\text{est}(R))\ \text{est}(R) \sqsubseteq \text{union}\ \text{est}(R) \text{--- a minimum in each set, then a minimum of those; } R \text{ transitive}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{setify lax natural}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (T\ A))} \text{---} \boxed{Q} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{fusion at (17.3.5d)}$	

gen $N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q$ (17.3.5d)	
fusion: the condition for fusion in (17.3.5c)'s last step, used to derive a definition of Q .	
circuit	Hinze–Marsden
$F(N\ A)\ (N\ (E\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)$	
$\sqsubseteq FNE(T\ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T\ A) \text{---} \boxed{Q} \text{---} N(T\ A)$ $(7.13), \text{ then zip, trans, moves lax natural}$	

(7.13) is $F(1, \text{est}(R)) \alpha \sqsubseteq \text{cp } P(\alpha) \text{ est}(R)$, mon-thm71 at the map α with $\frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha)$: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.